

COMPRESSED AIR TREATMENT SYSTEMS • WRD SERIES

Refrigerated Compressed Air Dryers

High-Efficiency Non-Cycling Direct-Expansion Dryers Engineered for Extreme Tropical Ambient Climates

Engineered for Constant +3°C Pressure Dew Point (ISO 8573-1 Class 4): Protects downstream pneumatic machinery, automated assembly cylinders, textile autoconers, and laser cutting heads from liquid moisture, rust fouling, and line freeze-ups.

Win Equipments WRD Series utilizes an advanced 3-in-1 modular heat exchanger combining an air-to-air regenerator, direct-expansion evaporator, and moisture demister in a single compact housing. Pre-cooling incoming air with outgoing cold dry air lowers compressor power consumption by up to 40% while preventing distribution pipe sweating.



NON-CYCLING DIRECT-X

SAVE WATER & POWER

50°C MAX INLET AIR

ECO R134A / R407C

20–2000

CAPACITY (CFM)

+3 °C

PRESSURE DEW POINT

16 bar g

PEAK RATED PRESSURE

< 0.2 bar

PRESSURE DROP

CORE ENGINEERING ADVANTAGES

3-in-1 Aluminum Exchanger

Integrated air-to-air regenerator, evaporator, and mechanical demister reduces piping connections by 70% and minimizes line pressure loss below 0.2 bar.

Zero Air Loss Drainage

Electronic capacitive condensate drain valve exhausts accumulated water automatically without venting costly compressed air, saving ₹42,000+ annually.

Hot Gas Bypass Modulation

Automatic pressure-balanced hot gas injection maintains stable evaporator temperature across 0% to 100% load variations, preventing coil freeze-up without compressor cycling.

TARGET INDUSTRIAL APPLICATIONS

Textile Spinning & Weaving Looms

CNC Machining & Robotic Clamping

PET Blow Molding Lines

Automotive Clear-Coat Paint Booths

Food & Beverage Packaging

Pharmaceutical Cleanrooms

THERMODYNAMIC ARCHITECTURE & SYSTEM INTEGRATION

Thermodynamic Process Flow & Piping Layout

Optimized Compressor Room P&ID Layout with Integrated Pre-Separation and Dual In-Line Filtration

The Win Equipments WRD series implements a four-stage refrigeration cycle engineered for heavy tropical operating loads. Saturated air enters at temperatures up to 50°C and is progressively cooled to condense water vapor into large droplets, followed by mechanical cyclone separation and zero-loss discharge.

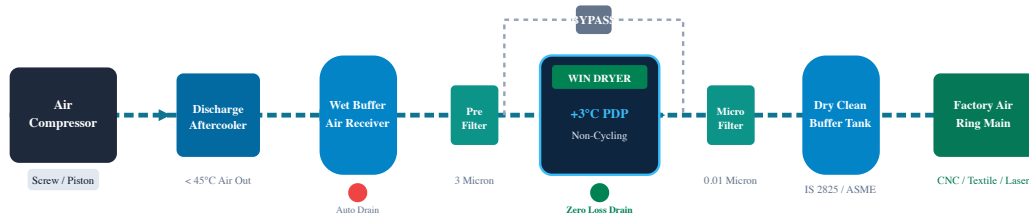


Figure 1: Standardized Win Equipments Compressor Room P&ID Flow Architecture

SUB-ASSEMBLY ENGINEERING & COMPONENT RIGOR

Copeland / Danfoss Compressors

Industrial-grade hermetic reciprocating and scroll compressors charged with eco-friendly R134a or R407C refrigerant. Fitted with internal thermal motor protection and vibration isolators.

Digital PID Temperature Panel

Front-panel digital controller provides real-time dew point readout, compressor operation status, auto-drain test switch, and alarm telemetry for refrigeration LP/HP cutouts.

Stainless Steel Demister

High-velocity centrifugal moisture separator with multi-vane stainless steel demister pad removes 99%+ of water aerosols, directing condensate directly into the drainage chamber.

Win Equipments 5-Stage Factory Acceptance Testing (FAT): Every WRD unit undergoes 1.5× hydrostatic vessel pressure testing, 24-hour dry nitrogen pressure holding test at 350 PSI, full-load electrical Megger testing, and dynamic dew point stability verification in our Arasur plant before final packaging and dispatch.

ENGINEERING SPECIFICATIONS MATRIX
WRD Series Technical Data (20 to 2,000 CFM)

 Standard Ratings: 7.0 kg/cm²(g) operating pressure, 45°C inlet air temperature, 35°C ambient temperature, +3°C PDP

MODEL NO.	NOMINAL CFM	AIR FLOW	POWER	SUPPLY VOLTAGE	CONNECTION	DIMENSIONS (L×W×H)	WEIGHT
Model Code	CFM	m ³ /hr	kW	V / Ph / Hz	Inlet / Outlet	mm	kg
WRD 20 S	20	34	0.45	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	38
WRD 30 S	30	51	0.45	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	40
WRD 40 S	40	68	0.45	230V / 1Ph / 50Hz	1/2" BSP	450 × 450 × 550	42
WRD 60 S	60	102	0.65	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	55
WRD 80 S	80	136	0.65	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	58
WRD 100 S	100	170	0.85	230V / 1Ph / 50Hz	1" BSP	500 × 550 × 750	60
WRD 125 S	125	212	0.85	230V / 1Ph / 50Hz	1.5" BSP	500 × 550 × 750	65
WRD 150 S	150	255	1.10	230V / 1Ph / 50Hz	1.5" BSP	600 × 650 × 900	90
WRD 200 S	200	340	1.25	230V / 1Ph / 50Hz	1.5" BSP	600 × 650 × 900	95
WRD 250 S	250	425	1.65	230V / 1Ph / 50Hz	2" BSP	600 × 650 × 900	105
WRD 300 S	300	510	1.65	230V / 1Ph / 50Hz	2" BSP	600 × 650 × 900	110
WRD 400 S	400	680	2.20	230V / 1Ph / 50Hz	2.5" BSP	800 × 800 × 1100	180
WRD 500 S	500	850	2.20	230V / 1Ph / 50Hz	2.5" BSP	800 × 800 × 1100	190
WRD 600 S	600	1020	3.50	415V / 3Ph / 50Hz	3" Flange	1200 × 850 × 1350	280
WRD 750 S	750	1275	3.50	415V / 3Ph / 50Hz	3" Flange	1200 × 850 × 1350	300
WRD 1000 S	1000	1700	5.50	415V / 3Ph / 50Hz	4" Flange	1500 × 1000 × 1500	450
WRD 1250 T	1250	2125	6.50	415V / 3Ph / 50Hz	4" Flange	1500 × 1000 × 1500	500
WRD 1600 T	1600	2720	8.50	415V / 3Ph / 50Hz	6" Flange	1800 × 1200 × 1600	650
WRD 2000 T	2000	3400	10.50	415V / 3Ph / 50Hz	6" Flange	1800 × 1200 × 1600	750

* Note: Model suffix 'S' indicates Single-Phase 230V or Single-Circuit architecture; suffix 'T' indicates Three-Phase 415V Dual-Circuit architecture. Custom models rated up to 40 bar g for PET blow molding available upon request.

ENGINEERING SELECTION & PLANT PROFILE

Thermodynamic Sizing & Factory Capabilities

Selection Formula, Operating Correction Multipliers & Arasur Plant Infrastructure

Air Dryer Sizing Formula (ISO 8573 Reference): Required Dryer CFM = Compressor CFM ÷ (Cp × Ct × Ca)

Pressure Factor (Cp)		Inlet Air Temp (Ct)		Ambient Temp (Ca)	
PRESSURE	CP	INLET TEMP	CT	AMBIENT	CA
4.0 bar	0.77	30°C	1.20	25°C	1.12
5.0 bar	0.86	35°C	1.10	30°C	1.06
6.0 bar	0.93	40°C	1.00	35°C	1.00
7.0 bar	1.00	45°C	0.85	40°C	0.90
10.0 bar	1.15	50°C	0.72	45°C	0.80

MANUFACTURING INFRASTRUCTURE & TRACK RECORD

Founded in 2008 by Mr. Ramasamy Ananthakumar, Win Equipments operates from its state-of-the-art registered manufacturing works in Arasur Post, Coimbatore on NH544. With 17+ years of engineering excellence, over 1,200+ industrial installations commissioned across India, and direct capital exports to Kuwait, Oman, and Jordan.

1,200+ COMMISSIONED PLANTS

GCC EXPORT CERTIFIED

ISO 9001:2015 QMS

DAC 50°C DESERT CERTIFIED



Request Technical Submittal & Commercial Quotation

Direct factory engineering consultation for custom capacity sizing, P&ID schematics & dimensional drawings.

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